

Using Messy Text in Future LLM Annotations

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Motivation

Introduction

- Researchers often inherit corpora and annotations from earlier projects.
- The annotations contains valuable expert human judgment.
- The texts and the annotations were *not* prepared for modern LLM workflows.

Can we re-use that prior coding work without recoding it?

Our Dataset

- News reports corpus of disappearances in Mexico (UMN / ODIM).
- 575 victims, 2,229 news reports across five states.
- Coded by trained human coders against a 7-section codebook.

The texts were scraped from the webpages. The codings were made at the

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Why the dataset is messy for downstream tasks

What that misalignment costs

- A victim's *method of capture* may live in one report, the *threats* in another: both annotated only at the victim level.
- Propagating victim-level labels to documents trains downstream models on wrong signals.

Why not just clean and re-code?

- Rule-based cleaning fails on per-site noise and on semantically related distractions.
- Manual re-cleaning and re-coding at this volume is slow, costly, and inconsistent.
- Many of the original sources are no longer retrievable from the webpages.

Contribution

Codebook attention

A turn-by-turn pipeline that uses the **codebook as the task schema** to focus the LLM:

1. **Extract** verbatim spans for each codebook field from each document.
2. **Summarize** the document into a structured prose anchored to those spans.
3. **Update** a per-victim summary across documents while preserving prior evidence.

What the pipeline produces

Output is clean, label-structured, and traceable back to source spans.

Method

Stage 1: Extraction of evidence

- For each input document and each codebook field: return verbatim text spans.
- Each extracted span is paired with a **character offset** into the source document.
- Output is a structured evidence record keyed by label.
- Every coded field is paired with the exact source phrase that supports it.

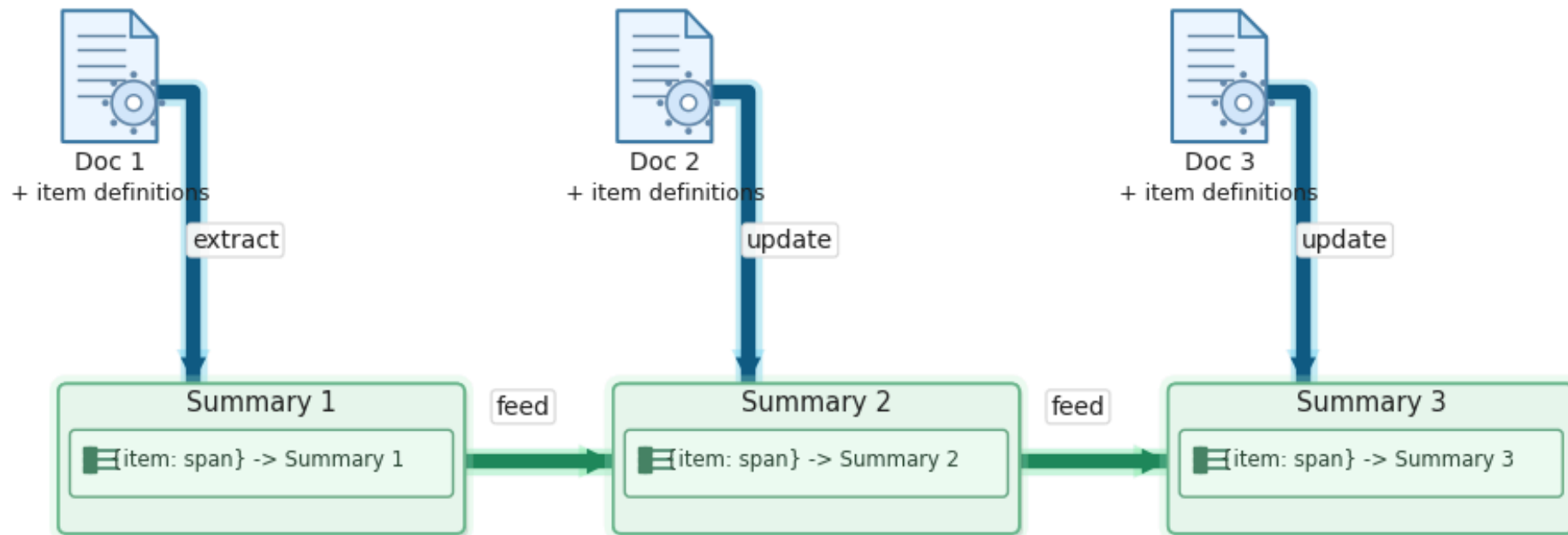
Stage 2: Summarization with given codebook taxonomy

- Input: document text + label definitions + extracted spans.
- The summary is written **field by field**, not as free paraphrase.
- Each sentence corresponds to one codebook label, anchored to its span.
- Result: the summary is easy for a downstream classifier to consume.

Stage 3: Stateful update across turns

- The pipeline keeps an explicit conversation state per victim.
- Each new document is checked against the current summary.
- Update only the fields where the new document adds information.
- If the document has no relevant content, the previous summary is preserved unchanged.
- Every turn logs: doc id, fields found, summary at that step.

Pipeline



Example: extracted spans

victim_id	label_key	span	offset
Veracruz_J J R R	desenlace	“...sigue en la búsqueda, se ha convertido en detective...”	2480
Veracruz_J J R R	vic_grupo_social	“Carmelo estudiaba arquitectura en una escuela privada de diseño...”	2691
N.L._Fabián H V	desenlace	“En el expediente no hay registros del plagio ni testimonios.”	2585
Veracruz_María Victoria C T	captura_tipo	“...sustraídos de un taxi entre la calle 13 y avenida 11, frente al DIF Municipal...”	1254
N.L._Fabián H V	captura_tipo	“...sustrajeron al empresario del vehículo siniestrado y se lo llevaron...”	4109

The offset is the character position where the span begins in the original text, so a researcher can trace it back and verify.

Example: turn-by-turn evolution

Coahuila_Ezequiel C T, four documents:

turn	info found	labels found	summary (excerpt)
0	:	:	(no summary produced)
1	yes	desenlace, vic_grupo_social	“Cuatro personas... fueron secuestrados por policías preventivos... y entregados a un cártel...”
2	:	:	(previous summary preserved unchanged)
3	yes	+ captura_tipo	“...siete jóvenes detenidos por policías locales y entregados a Los Zetas...”

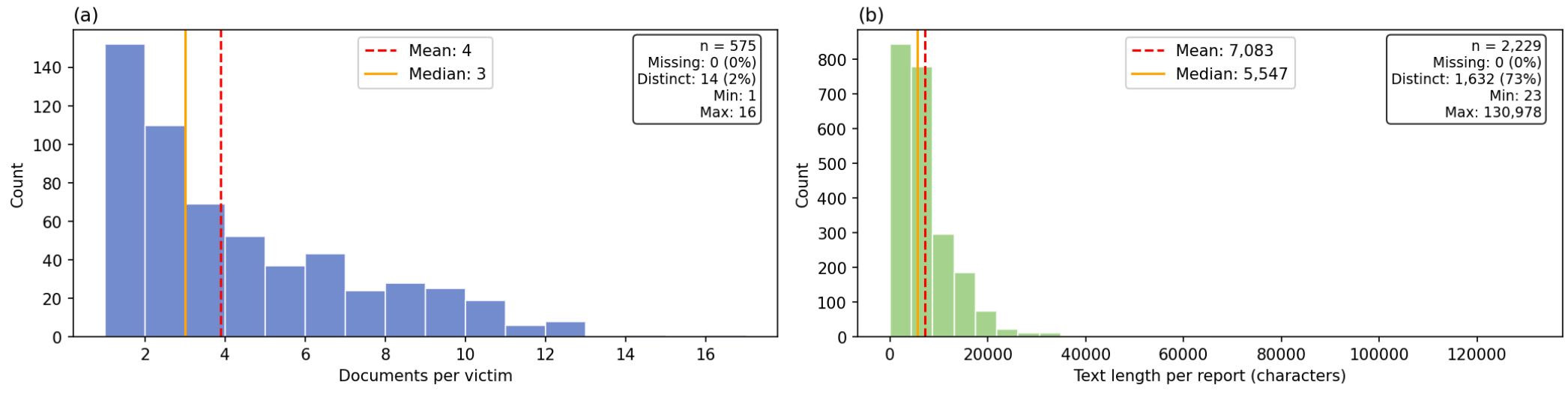
Noisy documents do not corrupt the running summary.
New evidence accumulates labels.

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Data

Press-based corpus

Document and Text Summaries



575 victims, 2,229 reports. Median 3 reports per victim. Median report length ~5,500 characters; long tail to >130k.

Three selected labels

Picking relevant and low missing rate labels:

- **Desenlace**: outcome of the disappearance (n = 393)
- **Captura tipo**: type of place of capture (n = 243)
- **Vic grupo social**: social group of the victim (n = 273)

Fine-grained categories were collapsed into broader groups before evaluation

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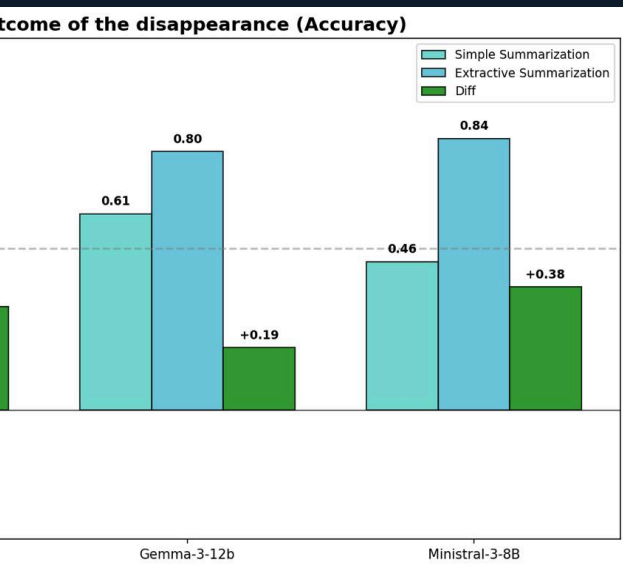
Evaluation

Why human annotations as the gold standard

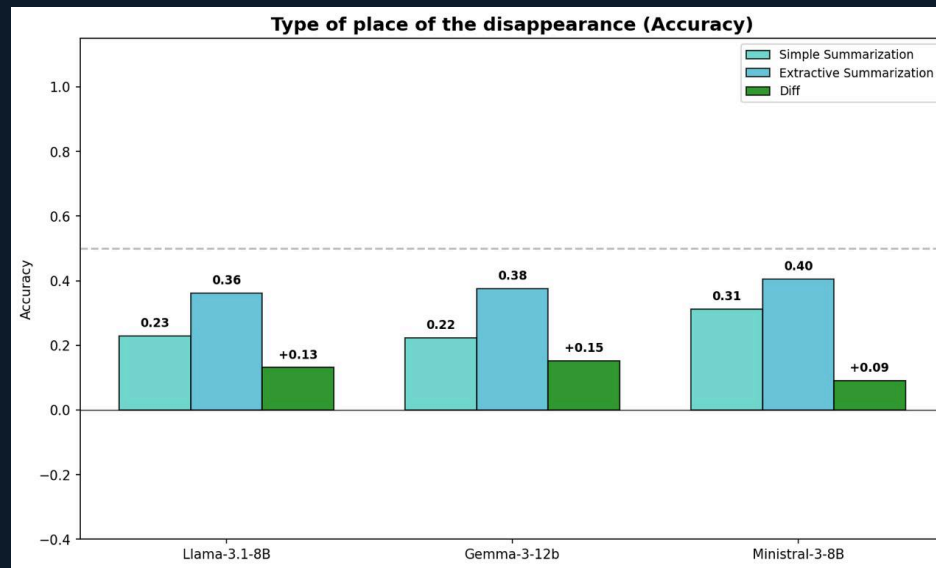
- Source-grounded LLM-as-judge methods (SummaC, G-Eval) affected by the noise.
- Reference required methods need gold standard summaries we do not have.
- The original human annotations encode the actual coding task.

Results

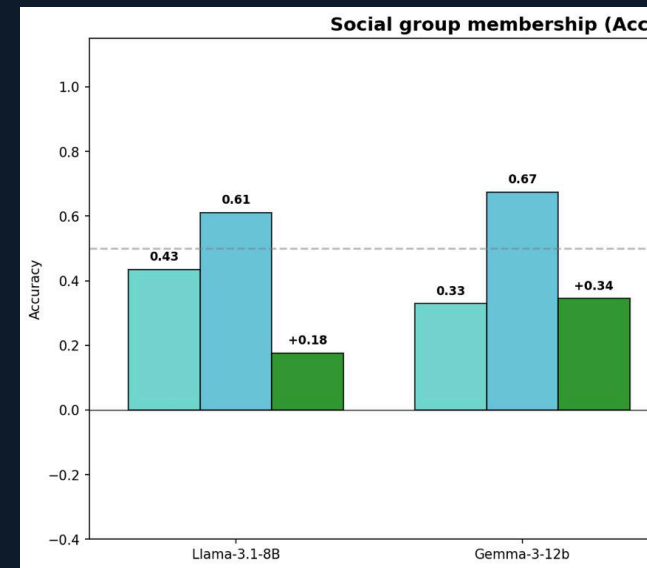
Simple summarization vs Extractive summarization



e (desenlace)



Place of capture



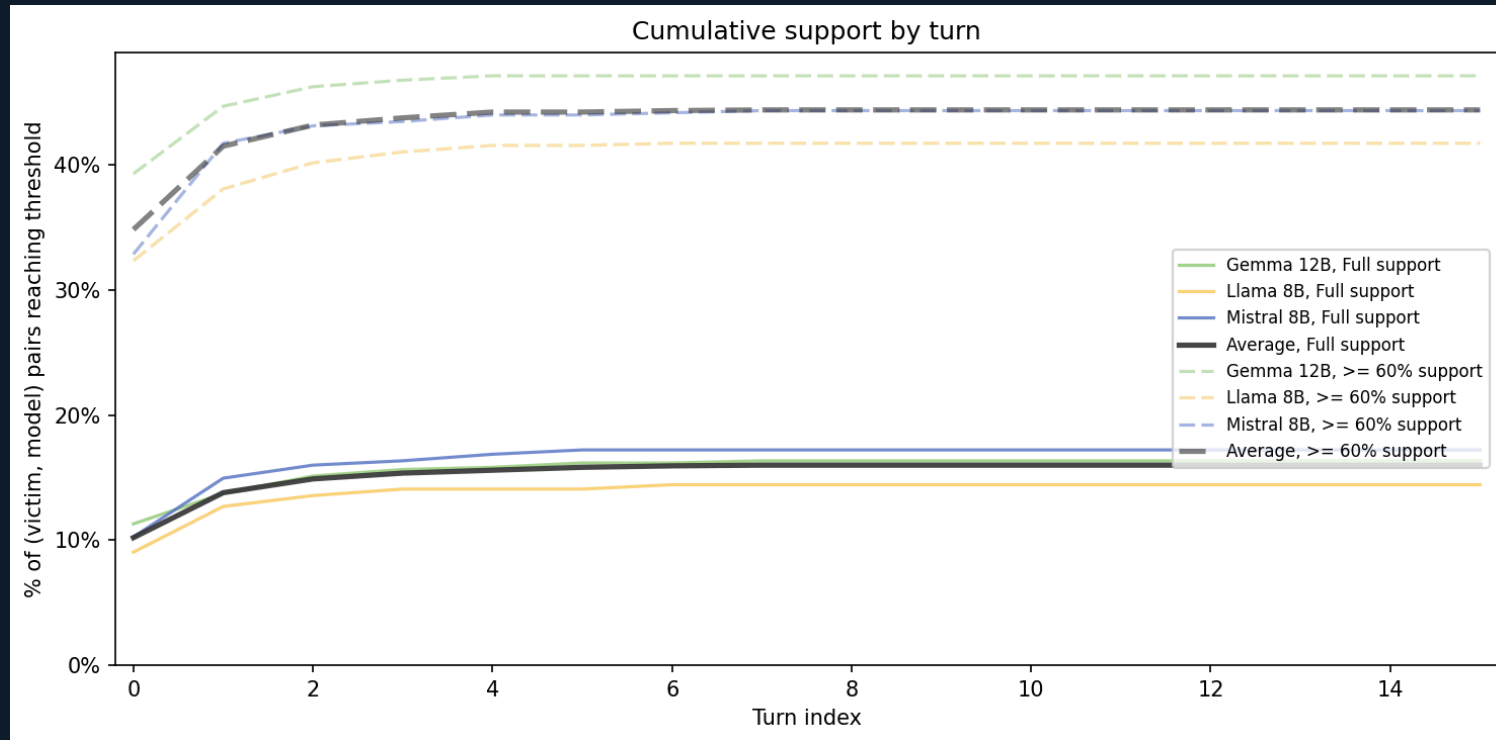
Social group

Cross-model comparison

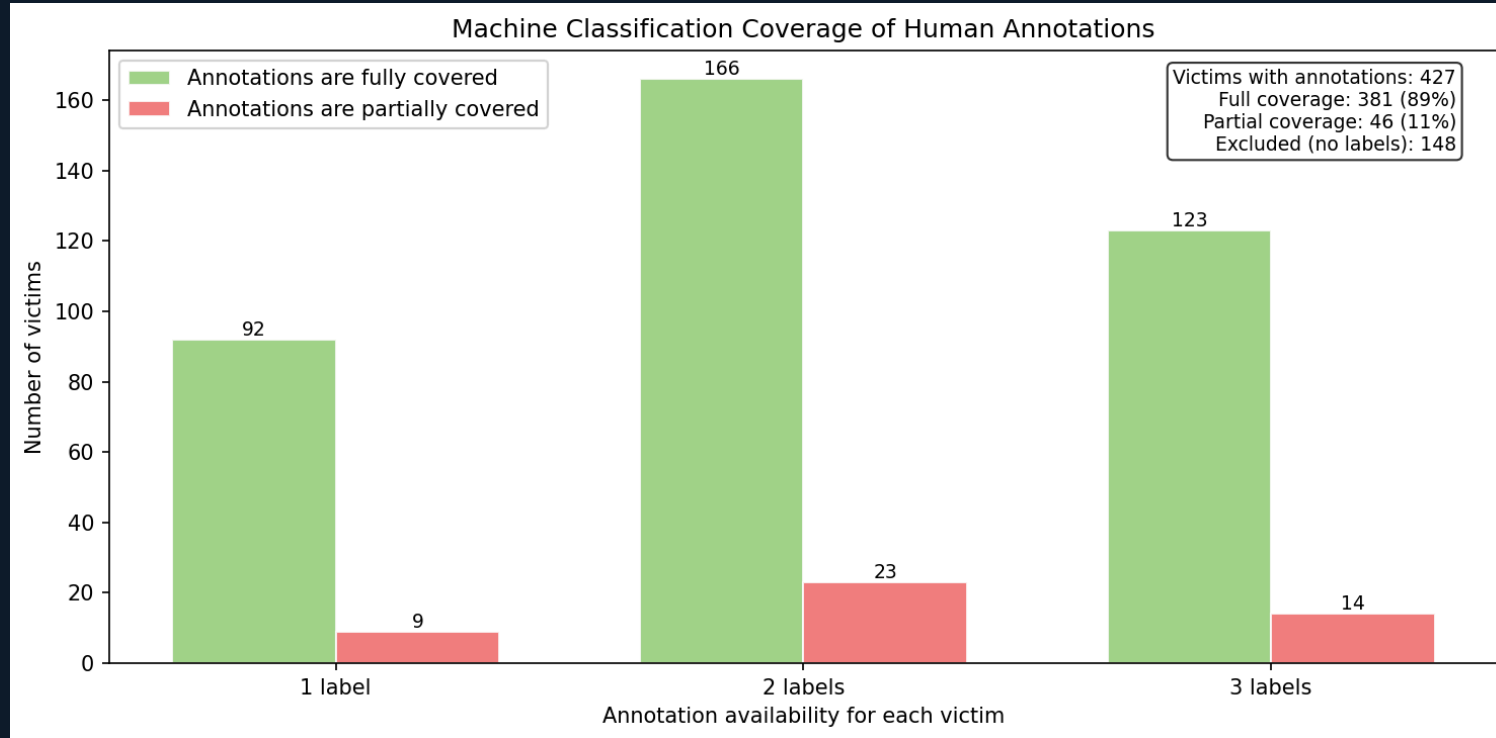
- All models benefit from the codebook-guided extraction.
 - Ministral 8B: best on outcome (0.84) and place of capture (0.40).
 - Gemma 12B: best on social group (0.67).
- The 4B parameter gap does *not* produce a consistent advantage.

The improvement comes from the workflow, not from any single model.

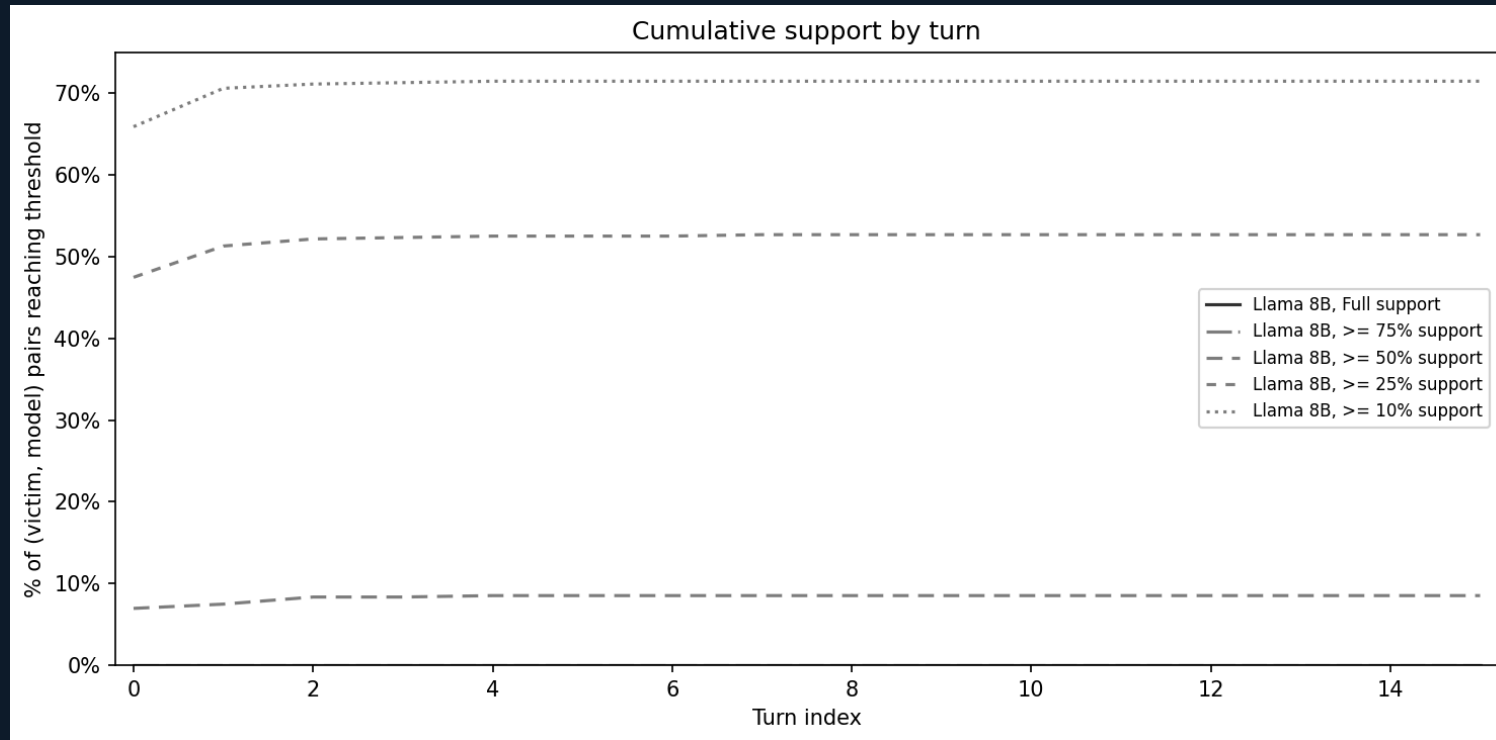
Recovering annotations by turns



Coverage by label availability



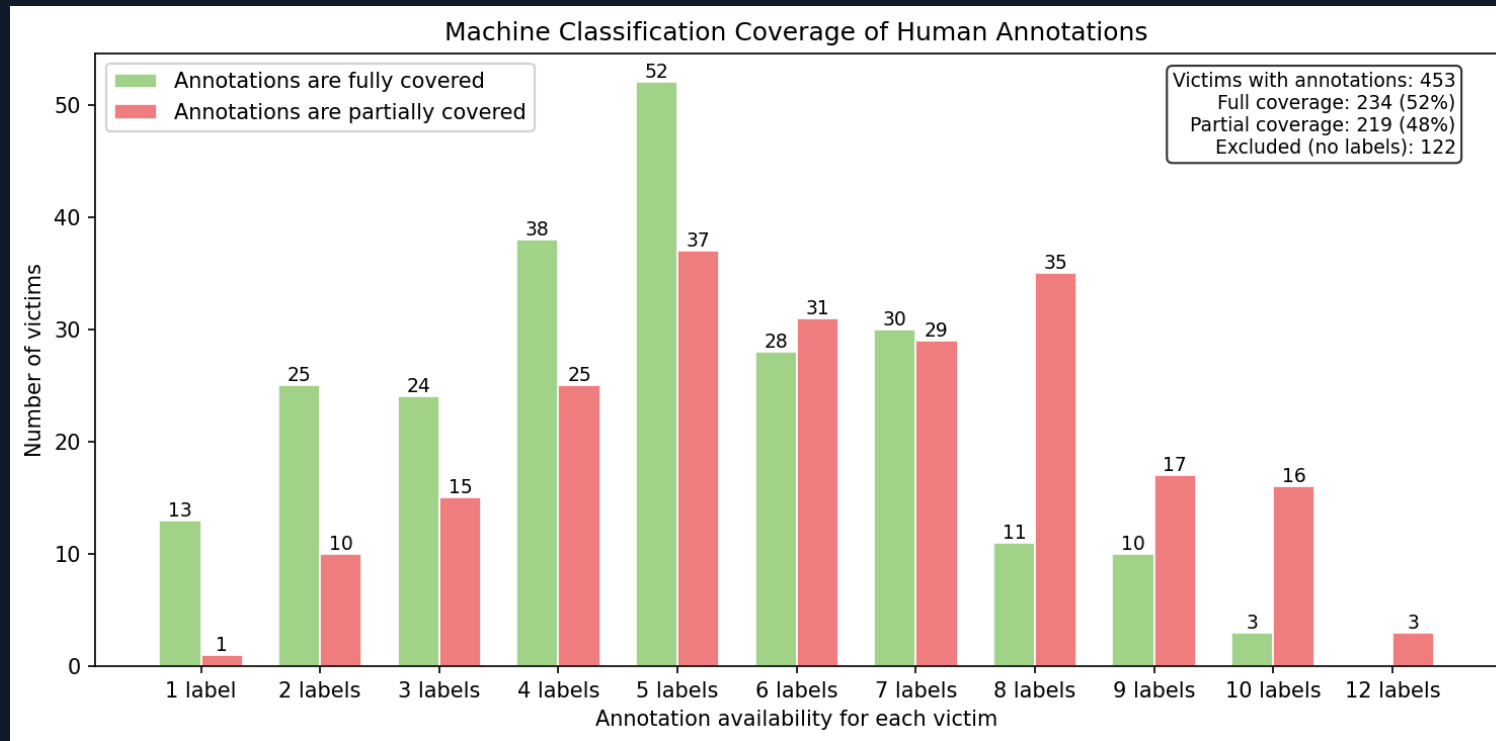
Recovering annotations by turns



Most victim can recover several labels after 2-3 turns; but more labels recovery

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Coverage by label availability



Raw text vs Extractive summary

Label	Input	Macro F1
Outcome	Raw text	0.656
	Summary	0.723
Social group	Raw text	0.418
	Summary	0.484
Place of capture	Raw text	0.600
	Summary	0.587

When using paraphrase-multilingual-MiniLM-L12-v2 with logistic regression classifier, the cleaned summary is favored compared to using the raw text.

Discussion

Compute vs human coding

Model	Time	GPU
Llama-3.1-8B-Instruct-AWQ-INT4	~5 h 27 min	A100
Ministral-3-8B-Instruct-2512	~51 min	A100
Gemma-3-12B-IT-INT4-AWQ	~38 min	A100

The pipeline processes the same corpus in hours, with a full turn-level audit trail. This shifts scarce expert labor from cleanup to validation, ambiguity

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Portability to other legacy corpora

- The codebook, label set, and prompts are configurable.
- Adapting to a new corpus requires swapping the schema, not rebuilding the pipeline.

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Backup slides

Model setup and runtime

Model	Time (full corpus)	GPU
Llama 3.1 8B Instruct (AWQ INT4)	~5 h 27 min	A100
Ministral 3 8B Instruct	~51 min	A100
Gemma 3 12B IT (AWQ INT4)	~38 min	A100

The Llama-8B server runs vLLM's legacy AWQ kernel (4× INT4 speedup only at batch size 1). Gemma's [awq_marlin](#) and Ministral's native FP8 path stay near peak throughput at our concurrency.

Selected labels after merging

Category	Count
Desenlace (n = 393, 32% missing, 2 distinct)	
Still disappeared	254
Found	139
Captura tipo (n = 243, 58% missing, 2 distinct)	
Places related to the victim	81
Public and institutional spaces	162
Vic grupo social (n = 273, 53% missing, 5 distinct)	
Professionals	145
Civil servants	61
Students	45
Other	19

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Related works

- **Non-LLM cleaning** (rule-based, OpenRefine): handles structure, fails on per-site layouts and semantic distractors.
- **LLM data prep** (cleaning, entity matching, classification): related capability, not source-grounded for our setting.
- **Imputation** (e.g. LLM-Forest): wrong tool: we want to *preserve* what's there, not infer what isn't.
- **LLM summarization**: usually assumes clean inputs and aligned units of analysis.
- **Closest application: Ventana a la Verdad** [@Sokol_etal2025]: RAG chatbot over the Colombian Truth Commission, optimized for QA, not codebook-aligned summaries.

Why not LLM-as-judge?

- **SummaC** scores source $\leftrightarrow$ summary consistency via NLI. Rewards keeping noise.
- **G-Eval** uses an LLM judge with structured rubrics. Same exposure to source-side noise.
- Either method may *penalize* a clean, focused summary for omitting distractive content.
- Our annotations are the only reliable anchor.

Hallucination evaluation, briefly

- Most relevant risk: **input-conflicting hallucination** (output not supported by source).
- Standard methods (FActScore, SummaC, G-Eval) assume clean sources or external evidence retrieval.
- The pipeline's design itself constrains hallucination: extraction step pins each label to a verbatim span with offset.

Three sources of error

1. **Document-level noise**: error pages, navigation, distractive content. Codebook-guided extraction filters most of this at extraction time.
2. **Lost documents**: some sources the human coders consulted are no longer retrievable. The pipeline records and quantifies this rather than hiding it.
3. **Label ambiguity**: some texts defensibly support more than one category. Counted as a classification error, but not a pipeline failure.

Knowing which is which is itself a contribution to transparency.

Sample messy reports

1. HTTP error page

Página no encontrada - Agencia de Noticias IRZA Saltar al contenido Agencia de Noticias IRZA desde 1992 Menú INICIO GUERRERO CHILPANCINGO ACAPULCO COSTA CHICA COSTA GRANDE CENTRO MONTAÑA NORTE TIERRA CALIENTE SIERRA CONGRESO GOBIERNO DEL ESTADO POLÍTICA POLICIACA DEPORTES NOTA ROJA × Buscar: Botón de búsqueda ¡Vaya! No se puede encontrar esa página. No hemos encontrado nada aquí. ¿Qué tal si pruebas el buscador? 404 error Buscar: Botón de búsqueda Gobierno del estado GOBIERNO DEL ESTADO Evelyn Salgado cumple con la salud de la Montaña y la Sierra entregan ambulancias a Tlapa y Leonardo Bravo 12 de septiembre de 2025 0 Facebook Twitter Email Compartir Chilpancingo, Gro.- En un acto de colaboración y compromiso social con la salud de las familias de Guerrero, el subsecretario de Desarrollo Político y Leer más Refuerza el gobierno de Guerrero comunicación y cooperación con autoridades fronterizas de los Estados Unidos 12 de septiembre de 2025 0 Evalúa Evelyn Salgado condiciones meteorológicas, fortalece cultura de la prevención en Guerrero y alista participación en Simulacro Nacional 12 de septiembre de 2025 0 El Bulevar de Las Naciones se ilumina para celebrar las fiestas patrias 12 de septiembre de 2025 0 Guerrero, el destino para disfrutar este fin de semana patrio 12 de septiembre de 2025 0 Se aleja Depresión Tropical Trece-E de Guerrero; Acapulco tendrá buen clima desde esta noche para celebrar las fiestas patrias 12 de septiembre de 2025 0 Guerrero pone orden en permisos y licencias; se blinda con tecnología contra el robo de autos: Evelyn Salgado 10 de septiembre de 2025 0 Reducirán el uso de plásticos de un solo uso en oficinas del sector educativo: SEG 10 de septiembre de 2025 0 Congreso CONGRESO POLÍTICA No hay distanciamiento entre Morena y el PT en el Congreso local: Urióstegui 10 de septiembre de 2025 0 Facebook Twitter Email Compartir Fernando Polanco Ochoa/IRZACHilpancingo, Gro.- El coordinador del Grupo Parlamentario de Morena del Congreso local, Jesús Urióstegui García, rechazó un

2. Website components mixed with body text

5/10/2019

Liberan a Matataxistas para ejecutarle orden de aprehensión | INFO7



- [SERVICIOS](#) INFO 7
 - [NOTICIAS](#)
 - [REPORTE CIUDADANO](#)
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 - [ESPECTÁCULOS](#)
 - [VIDA Y ESTILO](#)
 - [EDITORIAL](#)

3. Topically adjacent but unrelated content

Entradas relacionadas



Nuevo ataque a balazos contra policías: hay tres heridos



“Desapareció” joven empresario de Piedras Negras; piden ayuda para encontrarlo



“Piratea” y se adjudica Javier Guerrero fotos de eventos de Miguel Riquelme

← Concentran en Saltillo a 4 elementos de
Fuerza Coahuila por trata de migrantes

Jóvenes son presente y futuro, pero hay que
saberlos apuntalar: Martha Garay →

Ethics

- Victim names are masked while preserving dates, locations, and violation types.
- Cases involve ongoing risk to families from criminal organizations and state actors.
- Anonymization follows standard human rights documentation practice.
- Analytical value (patterns, language, framing) is preserved; identifying detail is not.

Pipeline I/O schema (for reference)

```
1 {  
2   "previous_summary": "<text>",  
3   "input_text": "<document text>",  
4   "label_definitions": {  
5     "captura_tipo": "Where was the victim taken from?",  
6     "desenlace": "What was the outcome of the disappearance?"  
7   }  
8 }
```

```
1 {  
2   "spans_by_item": {  
3     "captura_tipo": "...sustra\u00eddos de un taxi entre la calle 13...",  
4     "desenlace": "..."  
5   },  
6   "summary": "<label-structured text>"  
7 }
```